

The Metonym Game: An LLM Benchmark Without Ground Truth That Rises With the Models It Measures

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Paper under double-blind review

Abstract

We present evidence that analogy is at the core of LLM intelligence: a benchmark made of nothing but analogy-making and its judgement reproduces GPQA Diamond’s ranking of twelve models at $r = 0.98$ over three pooled runs. In the metonym game, LLMs compete in generating sets of analogous statements and rate each other’s sets on their own understandings of factual correctness, beauty, intelligence, distinctness, length, and structural diversity. Nothing enters from outside: every item is generated in play, and the scores come from the players’ ratings alone. Ground truth is replaced by the SVD of the factual rating matrix, whose left and right singular vectors rate the players as judges and as generators, two distinct ratings from one factorisation — to our knowledge a first for an LLM council-of-peers. For subjective criteria like beauty, judges are weighted by their rating consistency. Generating and judging are different skills: some strong generators are average judges. GPQA Diamond — difficult multiple-choice questions written by human experts — could not be more different in method, yet the two benchmarks correlate at Pearson $r = 0.98$, 95% CI [0.95, 0.99]; no leakage could be found. A council of the five best issues the official ratings; its contestable seats let the benchmark scale to any number of players and rise with the models it measures — a candidate steering signal for self-improving AI. Every number recomputes from a released package.

1 Introduction

Nearly every benchmark for machine intelligence needs a predetermined ground truth — golden keys and labels, oracle models, human panels. The benchmark reported here needs none of that. It is a game where frontier language models compete in making up analogies and then grade one another, and that grading is the single source of every score: no human raters, no answer key, nothing to look up.

The test is the metonym game. A player authors, from nothing, a context template — a paragraph of fixed wording with open slots — together with the sets of keywords that fill it, each set instantiating the template as a factually true description of a different domain; the keywords in corresponding slots are metonyms, metaphorically synonymous, and a set of them a metonym set. Table 1 shows one, written by a player. Its instantiations are each other’s metaphors and, as a set, parallel contexts: children of a common archetypal context, the abstract structure they share, of which the template is the literal representation. A long tradition treats seeing one structure across wildly different domains as central to thought and tests whether you recognise it; the game tests whether you can build it.

Because every item is produced fresh in the run, no fixed test set can leak into training; because correctness is settled sentence by sentence, the players’ own verdicts suffice — one matrix of their factual ratings reveals which judges are competent, with no labels at all (§3.3), and that subset is seated as the council that grades everyone. The canonical twelve-model run (§4) finds that judgement is the bottleneck — on this roster the strongest generators are middling judges — and that the key-free total tracks GPQA Diamond at Pearson $r = 0.98$, audited for a leak and found clean.

Contributions. (i) A production task for analogy that is falsifiable sentence by sentence, hence scorable without a key. (ii) A two-sided spectral estimator: one SVD of the self-produced factual rating matrix reads evaluator competence off the left singular vector and generator factuality off the right. (iii) A key-free reliability gate for subjective criteria: invariance under a sweep of the calibration anchor. (iv) A self-administering council with contestable seats. (v) The generation–judgement dissociation, and the $r = 0.98$ replication of a keyed benchmark by a key-free one, audited.

2 The metonym game

2.1 The machinery

An archetypal context is the cross-domain isomorphism General Systems Theory studies (von Bertalanffy, 1968).

Table 1 is one archetype as a player wrote it — the first of the submission that became the run’s anchor (§4.1).

Table 1: The anchor’s first archetype (§4.1), as a player wrote it. (a) The context template. (b) Its metonym table: MEMORY is realised as a bacterium’s methylation state, a climber’s route memory, a professional’s experience, an optimiser’s momentum term and an ant’s path integration — five mechanisms that are metaphorically synonymous in the archetypal context — metonyms. (c) Its first column played, first sentence of each form.

A [NAVIGATOR] moves through a [SPACE] by sensing local [GRADIENT] and adjusting its [TRAJECTORY] accordingly. The [NAVIGATOR] cannot perceive the entire [SPACE] at once; it relies on [SENSOR] that detect changes in [SIGNAL] concentration or intensity. When [GRADIENT] are steep and consistent, the [NAVIGATOR] converges efficiently toward [ATTRACTOR]. When [GRADIENT] are shallow, noisy, or conflicting, the [NAVIGATOR] may stall, oscillate, or become trapped in local [ATTRACTOR]. [INTERFERENCE] can distort the [GRADIENT], causing the [NAVIGATOR] to veer off course. Successful navigation requires not only sensitive [SENSOR] but also [MEMORY] of recent [TRAJECTORY] to distinguish genuine [GRADIENT] from transient [NOISE]. Some [NAVIGATOR] emit their own [SIGNAL] to recruit other [NAVIGATOR] toward the same [ATTRACTOR], creating collective [TRAJECTORY] that amplify the original [GRADIENT].					
[SLOT]	Bacterial Chemotaxis	Mountain Climbing	Career Development	Gradient Descent	Ant Foraging
NAVIGATOR	bacterium	climber	professional	optimizer	ant
SPACE	chemical environment	mountain	job market	loss landscape	terrain
GRADIENT	chemical gradient	slope	opportunity gradient	gradient	pheromone trail
TRAJECTORY	swimming path	route	career path	parameter update	foraging path
SENSOR	chemoreceptor	proprioception	network contact	backpropagation	antenna
SIGNAL	chemoattractant	elevation	opportunity signal	loss value	pheromone
ATTRACTOR	nutrient source	summit	desirable position	minimum	food source
INTERFERENCE	toxin	fog	misinformation	noisy data	rain
MEMORY	methylation state	route memory	experience	momentum	path integration
NOISE	Brownian motion	wind	market volatility	stochastic noise	environmental noise
Form (a), instantiation					
A BACTERIUM moves through a CHEMICAL ENVIRONMENT by sensing local CHEMICAL GRADIENTS and adjusting its SWIMMING PATH accordingly.					
Form (b), idiomatic rewrite					
Bacteria navigate chemical environments by detecting local concentration gradients and modulating their run-and-tumble behavior accordingly.					

Each parallel context is played in two forms: the instantiation (Form a), the mechanical substitution — only the slots filled, every other word carried over — which is what the factual grading acts on, because it must come out true sentence by sentence; and the idiomatic rewrite (Form b) in the target domain’s own register, showing the claim is not an artefact of the template’s phrasing. One template, mechanically swappable metonyms, true sentence by

sentence across maximal domain distance: that is what makes a metonym game decidable, and therefore measurable.

2.2 The rules

The game has N players and a non-competing administrator. Generation: a player creates archetypal contexts from scratch — a portfolio of K templates, M metonym sets each (five and five here), with instantiation and rewrite for every set. Evaluation: a player scores other players’ submissions on the rubric axes (§3.2) against one fixed reference submission pinned at an anchor value. A pass yields submission ratings for each portfolio and evaluator ratings for the judges: how well one detects the factual errors the other players collectively flag (factual competence), and how stable a standard it holds when the reference is re-pinned (rating consistency, §3.3). Each act is itself rated, so the framework is fully self-contained: no human raters, no external key.

3 The metonym game as a benchmark

3.1 Participants and protocol

Twelve frontier LLMs from Anthropic, Google and OpenAI are the participants (named in §4.4), each simultaneously generator and evaluator. The roster spans an order of magnitude in scale, three vendors, and adjacent versions within families; it is the models at hand, not a census. All twelve are called with Temperature 0, reasoning disabled, tools disabled, so the one greedy response is the measurement. Each model generates one portfolio — five archetypal contexts, each a template (typically 5–8 sentences, 6–10 slots; the prompt fixes only the counts, Appendix B) with a metonym table of five domains, 25 instantiations — then evaluates every other model’s portfolio under the six-axis rubric of Table 2, 1–10, one anonymised target per call alongside a fixed anchor portfolio pinned at 7 on every axis: a 12×11 evaluator-by-generator matrix (Appendix A). Every official rating pools three full runs of the game (§4.6); the anchor sweep behind the consistency ratings was run once, on the first run.

3.2 The rubric and the anchor

Table 2: The six-axis rubric, in the words the evaluator sees (Appendix B). No definition of beauty or intelligence is supplied; each judge rates on its own understanding.

Axis	Unit	The criterion as put to the evaluator
factual_per_pc	parallel context	each sentence is factually correct
beauty	archetype	beauty
intelligence	archetype	intelligence
instantiation_distinctness	archetype	the parallel contexts span very different domains; metonyms are far from synonymous
impressive_length	archetype	the archetypal template has impressive length
structural_diversity	portfolio	the archetypal contexts have very different system structures

Three design choices justify themselves on first principles. A fixed anchor: cardinal scores drift between evaluators — one model’s “8” is another’s “6” — and a reference pinned at a known score turns each idiosyncratic scale into a common one and recovers discriminability at the top, where the 1–10 ceiling compresses the strongest portfolios (§4.1). Holistic axes, minimally prescribed: a detailed rubric would leak back into generation as a template-construction tutorial, and we want to score what models recognise as beautiful or intelligent. impressive_length counterweights per-sentence factual scoring: without it the minimal tem-

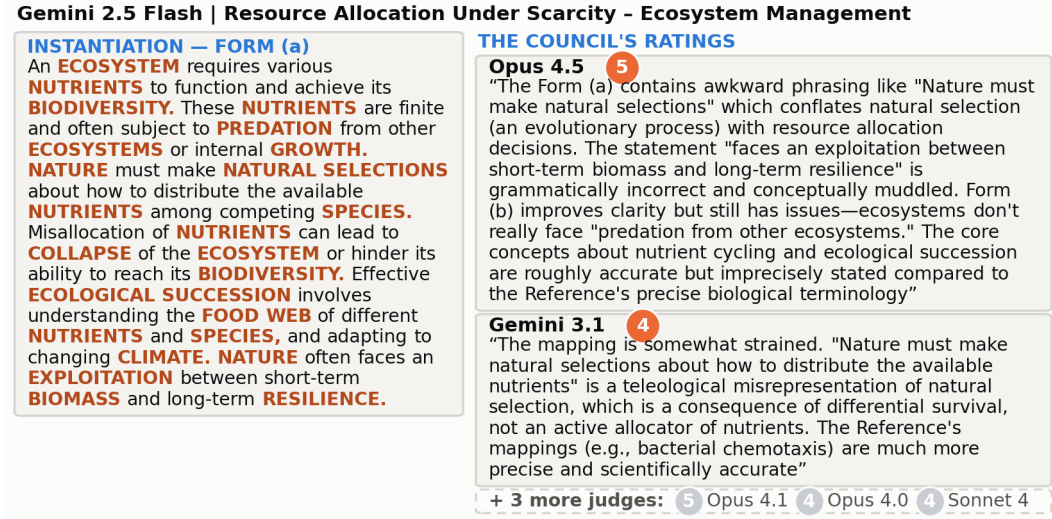


Figure 1: One parallel context under evaluation: the instantiation with its metonyms marked, and two of the five judges with their justifications (all five, the rewrite and the administrator’s synthesis in Appendix C). The “Reference” is the anchor, pinned at 7. All five judges isolate the same clause; the disagreement that remains, 4 versus 5, is about severity.

plate wins, and padding costs, since every added sentence is another claim to score. One evaluation, shown whole, is Figure 1.

3.3 Two key-free estimators

Factual competence — peer centrality. We assume that good evaluators agree with one another about which instantiations are factually weaker, once each evaluator’s own leniency is removed — the better two evaluators are, the more they agree. Stack the participants’ factual scores into one matrix — twelve evaluators against the 275 parallel contexts of the eleven scored portfolios — each entry the 1–10 rating used directly, row-centre it to remove each evaluator’s leniency, and take its SVD; the row-centred $\tilde{F}$ (evaluators $\times$ instantiations) is well approximated by its leading rank-one factor,

$$\tilde{F}_{sj} \approx \sigma_1 u_s v_j. \quad (1)$$

An evaluator’s rating tracks the consensus in proportion to its competence u_s times the instantiation’s factual standing v_j : competence and standing fall out of one factorisation, with no answer key. The left singular vector u is each evaluator’s factual-competence loading f — high when its ratings align with the participants’ shared signal, ≈ 0 when it rates everything alike or idiosyncratically — and, rescaled so the anchor model reads 7, $E^F = 7f/f_a$; the right vector, aggregated per generator, is G^F (Appendix A.2). f is each judge’s eigenvector centrality (Bonacich, 1972) in the leniency-removed agreement network, after clamping: an evaluator’s competence is its rating by the other evaluators, each weighted by its own competence — the highly trusted among the highly trusted. The construction is a graded relative of the classical label-free aggregators (Dawid & Skene, 1979; Parisi et al., 2014), which need categorical verdicts; discretising here would flip the marginal council seat.

Rating consistency — the anchor sweep. A reliable evaluator also needs a stable internal standard for each non-factual criterion. We sweep the anchor across 5, 6, 7 and 8 — the only difference between the four runs — and, per evaluator and axis, correlate (Pearson) the scores at one anchor with those at another, averaged over the six pairs, leave-self-out.

Re-pinning the anchor recalibrates the scale, not the rubric, and Pearson ignores a common shift or stretch; any reordering that follows a recalibration is not a change of judgement but flimsiness. Rescaled so the anchor model reads 7, consistency becomes an evaluator competence E^C on the generator’s scale.

Neither estimator can do the other’s job (Appendix A.8): peer centrality is licensed only where the one thing competent judges share is the truth — on taste, agreement is shared convention, and weighting by it would launder conformity into competence — and consistency cannot certify truth. The council is therefore seated on the factual axis, with consistency as the accompanying bar.

3.4 Council, total rating, and scaling by addition

The council. The council is the five players with the highest total T (eq. 2) and issues every official rating. The top five are the better factual judges (loadings 0.26–0.61, §4.2); the sixth reads 0.18; the line is drawn there; nothing prescribes five.

The total. Each side of the game splits into a factual and a criterion half: on generation, G^F (the SVD generation factuality) and G^C (the council’s leave-self-out mean over the five non-factual axes, each seat’s vote weighted by its own consistency on that axis); on evaluation, $E^F = 7f/f_a$ and $E^C = 7\bar{r}/\bar{r}_a$ — the anchor model scores 7 on every component. The total is the mean of the four, a symmetric 2×2 of {generator, evaluator} $\times$ {factual, criterion}:

$$T = \frac{1}{2}(G + E) = \frac{1}{4}(G^F + G^C + E^F + E^C). \quad (2)$$

Every rating carries a 95% percentile-bootstrap interval, E and T bootstrapped jointly (Appendix A.4).

Scaling by addition. A standing council scores any future model against the same anchor without re-deriving existing ratings. The seats are contestable — a contestant submits a portfolio, evaluates the incumbents’ portfolios and is scored by the seats under the definitions above, and wins a seat with a total T above the lowest seat’s by a margin the bootstrap can resolve. Because a contest convenes only the top of the field, two fixed ballast blocks — the weakest archived portfolios — join every contest’s graded set to keep the factual axis identified (Appendix A.6), and two guards: the spectral gap $\sigma_1/\sigma_2 \geq 2$ — the shared judgement standing clear of the strongest disagreement, sized in Appendix A.6 — and a spread of the seats’ E^F above 2.5 points. If either fails, no seat changes hands and the contest’s totals carry a caveat. The official leaderboard (§4.4) is issued on this contest basis. Contamination: items are generated fresh each run, so no fixed test set can leak into training. Every submission is archived; a new portfolio that copies an archived one is identified, and the contestant is asked for new templates — or the game is played in another language.

4 Results

4.1 Anchoring doubles resolution

The bootstrap opens with a raw pass — every portfolio scored by every other model with no anchor, averaged leave-self-out, 95% bootstrap intervals (2,000 resamples; Efron & Tibshirani, 1993). It supplies the baseline: the top-ranked portfolio, claude-opus-4.5’s, whose first archetype is Table 1, is pinned at 7 on every axis, leaving headroom above. Re-run anchored (Appendix G), the gap between a leading eight and a trailing four more than doubles relative to the spread of the means, while ranks within either band stay unresolved.

4.2 Evaluator factual competence

One SVD of the row-centred evaluator $\times$ instantiation matrix — no answer key — gives each evaluator a loading (Table 4, Appendix A.2; $\sigma_1/\sigma_2 = 2.2$, pooled). Five evaluators

— gemini-3.1-pro 0.61, claude-opus-4.5 0.52, claude-opus-4.0 0.35, claude-opus-4.1 0.35, gemini-2.5-flash 0.26 — stand above the rest.

Same-vendor robustness. A Claude-heavy evaluator set might read Claude-bloc agreement as truth. It does not: recomputing G^F with each vendor’s judges removed leaves the ordering essentially unchanged (Spearman ≥ 0.94 for every reduced set), gpt-4o-mini stays at the floor under every evaluator set, and a Claude-free set (Google + OpenAI judges) still places the Claude generators at the top (≥ 7.0).

4.3 Rating consistency and the council

The anchor sweep gives each evaluator a consistency on each axis (Table 5, Appendix A.3). The measure is self-consistency, not accuracy: gemini-2.5-flash (0.31) collapses on factual while its other axes hold. On the five subjective criteria generation and evaluation align (anchored cosine 0.85–0.92 per criterion, Appendix A.5) — the counterpoint to the factual axis, where they come apart (§4.4).

The council. The five seats — Gemini 3.1 Pro, Claude Opus 4.5, Gemini 2.5 Flash, Claude Opus 4.0 and Claude Opus 4.1 — clear both reliability bars (collapsed consistency $\bar{r}$ 0.94, 0.93, 0.81, 0.85, 0.87); Gemini 2.5 Flash is the weakest, its consistency interval [0.75, 0.86] straddling the bar.

4.4 The official leaderboard: judgement is the bottleneck

Every official number is council-issued on the contest basis of §3.4: each model is rated by the five seats — joined by the model itself when it holds no seat — over the incumbents’ portfolios, the two ballast submissions and its own, leave-self-out throughout. The two bases agree closely (Spearman 0.98; on the twelve-evaluator basis the fifth seat is a tie within 0.03).

Table 3: Final leaderboard — total T with its joint-bootstrap 95% CI, the evaluator half E , the generator half G , and the four anchored components; all council-issued against the fixed anchor, which reads 7.00 by construction. Adjacent ranks are resolved (non-overlapping intervals) only at 2–3 and 8–9; the rest are statistical ties. $E^F = 0.00$: no error signal in the model’s factual ratings.

Rank	Model	Council	T [95% CI]	E	G	G^F	G^C	E^F	E^C
1	★ claude-opus-4.5 (anchor)	council	7.00 [7.00, 7.00]	7.00	7.00	7.00	7.00	7.00	7.00
2	gemini-3.1-pro	council	6.92 [6.69, 7.24]	7.39	6.45	6.77	6.13	7.78	7.01
3	claude-opus-4.1	council	6.02 [5.71, 6.32]	5.06	6.98	6.94	7.02	3.65	6.47
4	claude-opus-4.0	council	5.81 [5.47, 6.12]	4.95	6.68	6.91	6.45	3.49	6.41
5	gemini-2.5-flash	council	5.75 [4.98, 6.21]	5.41	6.08	6.46	5.70	4.30	6.52
6	claude-sonnet-4	—	5.34 [5.09, 5.65]	4.10	6.58	6.95	6.21	2.07	6.13
7	gpt-4.1-mini	—	5.03 [4.37, 5.62]	4.31	5.74	6.23	5.25	1.66	6.97
8	gpt-4.1-2025-04-14	—	4.66 [4.49, 5.03]	3.50	5.82	6.59	5.04	0.77	6.23
9	gpt-4.1-nano	—	3.68 [3.31, 4.05]	3.35	4.02	4.55	3.48	0.26	6.43
10	gpt-4o-2024-08-06	—	3.34 [3.00, 3.56]	2.33	4.34	5.19	3.49	0.15	4.51
11	gpt-4o	—	2.95 [2.51, 3.29]	1.53	4.37	5.22	3.53	0.44	2.61
12	gpt-4o-mini	—	2.39 [1.96, 2.79]	1.17	3.62	3.71	3.53	-0.00	2.34

Two readings. The top five are the council: the five better factual judges of §4.2 are the five highest totals. Generation and evaluation do not coincide: the three Opus models, the strongest generators, are middling factual judges (E^F 3.5–3.7, the pinned anchor aside), while the strongest judge, Gemini 3.1 Pro ($E^F = 7.78$), generates mid-pack. Making a true claim and spotting a false one are different skills (West et al., 2024; Oh et al., 2024; Li et al., 2024), and rating them separately breaks the assumption behind key-free peer rankers (Ning et al., 2025; Zhang et al., 2025), which treat a strong generator as a strong judge.

5 Related work

As an intelligence test, the game probes the abstraction-and-analogy cluster a long tradition places at the centre of thinking (Gentner, 1983; Hofstadter & Sander, 2013; Penn et al., 2008; Chollet, 2019; Mitchell, 2021). The classical instruments (Srivastava et al., 2023; Webb et al., 2023; Lewis & Mitchell, 2024; Chollet, 2019) give source and target and ask for one selection or completion, scored against a key; the metonym game asks for many coupled slots across unrelated domains, built from scratch, and is the first to make analogical production falsifiable sentence by sentence.

As a self-contained method, the council sits in the unsupervised peer-evaluation line, which already removes the gold key: single-judge protocols (Zheng et al., 2023) trust one judge; PoLL (Verga et al., 2024) adds a panel but trusts it as given; LLM-as-Examiner (Bai et al., 2023) lets the examiner write the questions; PiCO (Ning et al., 2025) lets unlabelled models answer and grade one another and recovers an ability ordering from peer agreement alone, and UPME (Zhang et al., 2025) extends it to vision-language. We weight by agreement only where agreement is licensed to mean truth, and our council certifies and re-contests its own judges. Label-free spectral aggregation is one-sided in both its lineages: in the aggregation lineage (Parisi et al., 2014; Dawid & Skene, 1979) predictors classify a fixed external dataset, so there is no generator to score; in the reputation lineage (EigenTrust, Kamvar et al., 2003) EigenBench (Chang et al., 2026) has LLMs judge one another’s responses against a written value constitution and takes the leading eigenvector of a model-by-model trust matrix as each model’s score: one number is both standing and weight as a judge, applied to every criterion, subjective ones included. Our matrix is two-sided: one SVD scores judges on the left and generators on the right, the two are kept apart, and the generation–evaluation gap of §4.4 — which contradicts that premise on the factual axis — is definable only because the test is self-produced. Rating consistency applies the judge-reliability principle of invariance under non-semantic perturbation (Weng et al., 2026; Bellibatlu et al., 2026) to subjective, ground-truth-free criteria on self-produced items — to our knowledge a new use of the sweep. Don-Yehiya et al. (2026) find the anchor should be recalibrated to the field’s range — our rule — and warn against a top-model anchor; ours is pinned at 7 with headroom above (§4.1).

6 Discussion and limitations

Two yardsticks. The factual estimator’s one assumption — the only thing competent evaluators share is the truth — is what licenses agreement-weighting for facts. The disclosed alternative applies it to taste as well (an authority rating, one SVD per subjective axis; Appendix A.7); we decline it because on taste the dominant axis of agreement is shared convention, so weighting by it would reward the judge nearest the mean. Peer centrality’s weakness is the shared misunderstanding; rating consistency asks only whether a judge holds a firm standard.

A sustainable yardstick. When the field outgrows the anchor — a standing total above the pinned 7 by a resolvable margin — the anchor is replaced by the stronger submission and models are re-scored; the ballast stays fixed. Contestable seats keep the judges current, retesting the scale: the benchmark rises with the models it measures, which a keyed test cannot: its key-makers are fixed.

A hypothesis for the agreement. The total tracks GPQA at 0.98 (§4.5). GPQA is an accepted measure of capability with no theory of intelligence behind it: expert-written questions and a key (Rein et al., 2023). The game reproduces its ranking as closely as the measurement error of the two instruments allows (Appendix D.1), so whatever GPQA measures, the game measures too. Of the eight constructs the game demands, four are analogy under cognitive science’s own names (Appendix E): that is the sense in which analogy is at the core — sufficient by itself, not the only thing that could be. An agreement of 0.98 suggests that the two instruments measure the same thing, and their methods share nothing on the surface, so the common thing is in the depth. Our hypothesis is that a language model holds its knowledge as archetypal contexts, each instantiated in the topic

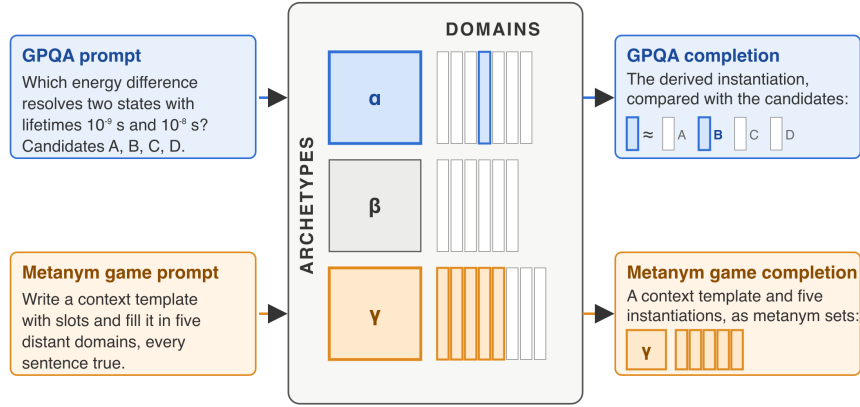


Figure 3: A hypothesis for the 0.98: the metanym game reproduces the way the model organises its knowledge, and GPQA runs on that organisation. An archetype (α , β , γ) is held once, about eight times the size of a domain’s instantiation (the slices); each domain adds only its metanym set. GPQA: archetype and domain are given; the model derives the instantiation and selects the matching candidate. The game: nothing is given; the model selects the archetype and domains that best satisfy the criteria and writes them out.

domains where it applies; the metanym game reproduces this organisation, and GPQA runs on it (Figure 3). Both tests are one retrieval followed by one selection. In GPQA the archetype and the domain are given by the question: the model derives the instantiation and selects the candidate that matches it, the mechanism in use. In the game nothing is given: the model selects the archetype and the domains that best satisfy the prompt’s criteria, factual truth among them, and writes them out, the mechanism itself. The hypothesis is untested here. The argument behind it, and what would refute it, is Appendix H. It predicts that the agreement is domain-matched, a model’s factual score on parallel contexts in a field tracking its GPQA accuracy in that field, and that archetype and topic domain are separable in a model’s latent space.

Steering signal, and its caveat. Self-improvement, the council governing its own rules, is specified but not exercised. A system optimised against T is optimised against a consensus it participates in, so gains can come from courting the consensus; the partial answers are the two quarters consensus does not own and independently constituted councils.

Scope. The runs share one configuration — one prompt template, one roster — so the bootstrap intervals measure item dispersion only (the four anchor values give the same leaderboard, Spearman 0.90–0.96); the leading group sits at its discrimination floor; a quarter of a non-council model’s total rests on the contest’s easier consistency test; no seat has yet been contested. Peer consensus is conservative against anomaly: a synthetic evaluator that reproduces the consensus and then inverts a fifth of its verdicts loses most of its competence (Appendix A.7), a penalty a dissenter pays on only half of T .

7 Conclusion

The metanym game is a structural test of intelligence built entirely of analogy, falsifiable sentence by sentence. The council-of-peers benchmark needs nothing outside itself: truth as the dominant axis of inter-evaluator agreement, reliability as invariance under a swept anchor, judges certified by the participants, contestable seats — and one external check, by design, $r = 0.98$ against GPQA Diamond.

AI use statement

Every original idea in this work is the author’s. The work was developed in a sustained dialogue with generative AI assistants (Anthropic’s Claude), directed by the author throughout. Within that dialogue the assistants criticised the theory, the estimators and the experimental design as a reviewer in the field would, and proposed hypotheses and experiments alongside the author’s; none was adopted without a competing hypothesis and a further experiment to test it. They implemented the analysis methods, cleaned and reformatted the evaluation records and GPQA response logs, and helped interpret results; they were also used for code, literature search, figures, references and first drafts, which the author rewrote into the author’s own text. No data and no proofs were generated by AI: every rating in the paper comes from the twelve rated models of §3.1. All AI-assisted output was checked against the deterministic re-derivation of every published number (Reproducibility statement). The author takes responsibility for the final content.

Ethics statement

The study evaluates commercial language models via their public APIs on self-generated material; no human subjects, personal data or annotators are involved. The benchmark is proposed as a candidate steering signal for self-improving systems. Steering by it would mean letting a consensus of models, with no human key, decide what counts as better, and a misunderstanding the models share would then be reinforced rather than corrected. The paper keeps the factual axis answerable to independent checks (§4.5) and leaves the self-improvement loop specified but unrun (§6); we regard that as the condition under which such a signal may be used. We consider the release of a key-free, contamination-resistant evaluation to be net positive for the field’s ability to measure models past the point where human-written keys remain reliable, and we release all data and code under permissive licences.

Reproducibility statement

Every number, table and figure in this paper recomputes deterministically from a released package (anonymised repository, supplementary material): the pinned evaluation runs (the un-anchored pass, the anchored pass at each of the four anchor values, and the three runs of the full pipeline — the canonical run and two regenerations), the raw per-question GPQA administration including both stages’ records and the first-pass log, the anchor and ballast portfolios, the verbatim generation and evaluation prompts (Appendix B), the analysis scripts, and a one-command reproduce.sh whose steps are each labelled by the exhibit they produce. The re-analysis makes no API calls and needs no credentials; it takes about a minute. The estimators are specified to the equation in Appendix A, including every convention (self-entry filling, row-centering, sign and clamp, Procrustes alignment of bootstrap replicates, the joint (submission, archetype) resampling grid). Producing a new run — re-querying the models — is deliberately not part of the package: it costs API budget and is non-deterministic by construction; §4.6 reports what moves across three independent regenerations.

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A Rating estimators

Every rating comes from one object: the scores the participants produce when each model grades the others' portfolios, swept across the anchor. No external answer key is used.

Panel, tasks, anchor. Twelve models, indexed $s, t \in \{1, \dots, 12\}$, are each a submission (its portfolio is graded) and an evaluator (it grades the others). Every model was asked to grade every portfolio including its own, and those self-evaluations are released; but no rating below uses a model's grade of its own portfolio — every estimator is leave-self-out. A portfolio holds five archetypes, each realised as five parallel contexts. Scoring uses six axes on a 1–10 scale: a factual axis (once per parallel context) and five non-factual axes — beauty, intelligence, instantiation-distinctness, impressive-length (once per archetype) and structural-diversity (once per portfolio). Every score is relative to the anchor — the model a whose portfolio won the un-anchored initial selection — declared to score 7 on every axis. The anchor value is swept, $\theta \in \Theta = \{5, 6, 7, 8\}$, and $\theta^* = 7$ is the production anchor. Write $r_{t,s,x,u}(\theta)$ for evaluator t 's score of unit u of axis x of submission s at anchor θ . A model earns a generation rating G and an evaluation rating E (A1); E has two parts (A12).

A.1 Generation rating

Evaluator t 's overall score of submission s averages within each axis, then across axes:

$$o_{t,s} = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \frac{1}{|U_x|} \sum_{u \in U_x} r_{t,s,x,u}(\theta^*), \quad (\text{A2})$$

and the six-axis generation rating is the leave-self-out mean over the council $\mathcal{C}$,

$$G_s = \frac{1}{|\mathcal{C} \setminus \{s\}|} \sum_{t \in \mathcal{C}, t \neq s} o_{t,s}. \quad (\text{A3})$$

Intervals: 95% percentile bootstrap over the per-(submission, archetype) units; a gap $G_s - G_{s'}$ is resolvable when the paired bootstrap puts its interval clear of 0. The official leaderboard's G is the split form (A13), $G = \frac{1}{2}(G^F + G^C)$, not (A3).

A.2 Factual competence and generation factuality (SVD)

Stack the factual scores into F (evaluators $\times$ instantiations), each entry the 1–10 rating used directly, no thresholding. Self-entries and the rare missing entries are set to the anchor value:

$$F_{sj} = r_{s,j} \in \{1, \dots, 10\} \quad (\text{self-entries} = \theta^* = 7). \quad (\text{A5})$$

Centre each row (subtract the evaluator's mean, removing its leniency) and keep the leading triple:

$$\tilde{F} = F - \bar{r} \mathbf{1}^\top = U \Sigma V^\top, \quad \sigma_1, \quad u \equiv U_{\cdot 1}, \quad v \equiv V_{\cdot 1}. \quad (\text{A6})$$

The left singular vector is factual competence,

$$f \equiv u, \quad \text{signed so } \sum_s f_s > 0, \quad f_s^+ = \max(f_s, 0), \quad (\text{A7})$$

clamped at zero so an evaluator anti-correlated with the consensus carries no weight. Centering is essential: raw scores cluster at the anchor, so on the un-centred matrix the leading axis is the shared level and ranks the most lenient evaluators highest. Equivalently u is the leading eigenvector of the row-centred inter-evaluator Gram $\tilde{F} \tilde{F}^\top$. Because every row of

$\tilde{F}$ sums to zero, v sums to zero and is oriented so that positive means factually stronger ($\text{corr}(v, \text{column means of } \tilde{F}) > 0$).

The competence-weighted consensus rating of instantiation j reads v back on the 1–10 scale,

$$\hat{r}_j = C + \kappa v_j, \quad C = \frac{\sum_s f_s^+ \bar{r}_s}{\sum_s f_s^+}, \quad \kappa = \sigma_1 \frac{\sum_s f_s^+ u_s}{\sum_s f_s^+} > 0, \quad (\text{A8})$$

the rank-one approximation of the competence-weighted mean rating (the two agree within 0.14 on the canonical run, $r = 1.00$). Averaging over a generator’s own instantiations J_g gives the key-free generation-factuality rating

$$G_g^F = \frac{1}{|J_g|} \sum_{j \in J_g} \hat{r}_j, \quad (\text{A9})$$

already on the 1–10 scale: clean instantiations sit at $v_j \approx 0$, hence at $C \approx 7$. Its interval resamples the generator’s own $\hat{r}_j$ with the consensus (C, κ, v) held fixed. The one assumption: the only thing the evaluators share is the truth — a same-vendor bloc with a common bias would add a spurious shared component, which is why competence is read off a vendor-diverse panel with the shared-bias check of §4.2.

Table 4: Evaluator factual competence E^F (left singular vector; “anchored” = $7f/f_a$) and generator factuality G^F (right vector, per generator), twelve-evaluator basis, three runs pooled (805 items). † loading interval includes zero; no interval printed.

Model	E^F loading	E^F anchored	95% CI	G^F	95% CI
gemini-3.1-pro	0.61	8.24	[7.33, 9.30]	6.78	[6.62, 6.89]
claude-opus-4.5	0.52	7.00	[7.00, 7.00]	7.00	(anchor)
claude-opus-4.0	0.35	4.69	[3.45, 6.04]	6.93	[6.89, 6.97]
claude-opus-4.1	0.35	4.65	[3.56, 5.96]	6.99	[6.91, 7.07]
gemini-2.5-flash	0.26	3.47	[1.62, 5.05]	6.49	[6.27, 6.68]
claude-sonnet-4	0.18	2.37	[1.53, 3.36]	6.96	[6.92, 7.00]
gpt-4.1-mini	0.10	1.34†	—	6.26	[6.07, 6.43]
gpt-4.1-2025-04-14	0.06	0.79	[0.39, 1.53]	6.56	[6.39, 6.69]
gpt-4o	0.04	0.47	[0.28, 0.70]	5.33	[5.12, 5.53]
gpt-4o-2024-08-06	0.03	0.35	[0.18, 0.61]	5.39	[5.16, 5.62]
gpt-4.1-nano	0.02	0.28†	—	4.82	[4.01, 5.38]
gpt-4o-mini	-0.00	-0.00†	—	4.14	[3.60, 4.69]

A.3 Rating consistency

For evaluator s and axis x , let $v_{s,x}(\theta)$ be s ’s axis- x scores across that axis’s units at anchor θ , leave-self-out (50 units for the four per-archetype axes, 250 for factual, 10 for structural diversity; 55 / 275 / 11 for the anchor, whose own portfolio is not among the graded eleven). The per-axis consistency is the mean Pearson correlation over the anchor pairs on which both vectors are non-constant,

$$r_{s,x} = \frac{1}{|\mathcal{P}_{s,x}|} \sum_{(\theta, \theta') \in \mathcal{P}_{s,x}} \text{corr}(v_{s,x}(\theta), v_{s,x}(\theta')). \quad (\text{A10})$$

The collapsed score that gates the council averages the four per-archetype non-factual axes into one value per (submission, archetype) before correlating,

$$\bar{r}_s = \frac{1}{|\mathcal{P}_s|} \sum_{(\theta, \theta') \in \mathcal{P}_s} \text{corr}(v_s(\theta), v_s(\theta')), \quad (\text{A11})$$

so axis-idiosyncratic noise partially cancels (gemini-2.5-flash: row mean of Table 5 ≈ 0.70 , collapsed 0.81). Factual is A.2’s job and is excluded; structural diversity, one score per portfolio, is too coarse and is excluded too. Design counts are met for eight of twelve evaluators; a missing grading call costs units, not correctness (dropped pairwise).

Table 5: Anchor-sweep consistency per evaluator and axis (A10): mean pairwise Pearson correlation of scores across anchor values 5/6/7/8. It measures self-consistency, not accuracy; the factual column is distinct from the factual-competence loading. gpt-4o-mini’s factual entry is undefined (near-zero variance). The initial council’s collapsed values $\bar{r}$ (A11) with 95% CIs: gemini-3.1-pro 0.94 [0.92, 0.96], claude-opus-4.5 0.93 [0.89, 0.96], gemini-2.5-flash 0.81 [0.75, 0.86], claude-opus-4.0 0.85 [0.80, 0.89], claude-opus-4.1 0.87 [0.82, 0.91]; their factual loadings with CIs: 0.58 [0.56, 0.60], 0.55 [0.54, 0.58], 0.37 [0.31, 0.40], 0.35 [0.32, 0.35], 0.28 [0.27, 0.30].

Evaluator	factual	beauty	intelligence	distinctness	length	struct
gemini-3.1-pro	0.88	0.83	0.90	0.87	0.90	0.89
claude-opus-4.5	0.90	0.86	0.85	0.84	0.91	0.85
claude-opus-4.1	0.83	0.84	0.85	0.72	0.84	0.92
claude-opus-4.0	0.78	0.83	0.84	0.69	0.83	0.86
gpt-4.1-mini	0.87	0.77	0.75	0.76	0.84	0.81
claude-sonnet-4	0.70	0.75	0.77	0.75	0.78	0.81
gpt-4.1-2025-04-14	0.24	0.64	0.73	0.74	0.82	0.84
gemini-2.5-flash	0.31	0.77	0.66	0.50	0.82	0.85
gpt-4.1-nano	0.65	0.63	0.64	0.44	0.37	0.36
gpt-4o-2024-08-06	0.59	0.58	0.55	0.38	0.53	0.69
gpt-4o	0.13	0.32	0.24	0.12	0.40	0.34
gpt-4o-mini	n/a	0.25	0.36	0.27	0.10	0.23

A.4 The council and the total

The sweep keeps the anchor interior to the 1–10 scale. At its ends, 5 and 8, the room on one side of the anchor shrinks, so part of the residual disagreement between anchor values is compression of the scale rather than reordering; the consistency values are conservative to that extent.

A model is reliable when its competence sits clear of the inert band — its 95% bootstrap interval on f separates it from the near-zero cluster — and $\bar{r}_s \geq 0.78$. Eight of twelve clear the consistency gate; five clear both, and that conjunction seats the council. Each evaluator index is rescaled so the anchor scores 7,

$$E_s^F = 7 \frac{f_s}{f_a}, \quad E_s^C = 7 \frac{\bar{r}_s}{\bar{r}_a}, \quad (\text{A12})$$

so the singular vector’s arbitrary scale cancels. Generation splits the same way: G^F (A9) and the criterion quality G^C , the council’s mean over the five non-factual axes, reliability-weighted — each judge’s vote on axis x weighted by its own $r_{t,x}$:

$$G_s^C = \frac{1}{|\mathcal{X}_5|} \sum_{x \in \mathcal{X}_5} \frac{\sum_{t \in \mathcal{C} \setminus \{s\}} \max(r_{t,x}, 0) \bar{r}_{t,s,x}}{\sum_{t \in \mathcal{C} \setminus \{s\}} \max(r_{t,x}, 0)}, \quad (\text{A12b})$$

with $\bar{r}_{t,s,x}$ judge t 's mean score of s on axis x at the production anchor. Then

$$G_s = \frac{1}{2}(G_s^F + G_s^C), \quad E_s = \frac{1}{2}(E_s^F + E_s^C), \quad (\text{A13})$$

$$T_s = \frac{1}{2}(G_s + E_s) = \frac{1}{4}(G_s^F + G_s^C + E_s^F + E_s^C). \quad (\text{A14})$$

Intervals. Every rating carries a 95% percentile bootstrap (10^3 – 10^4 replicates). Units: G — the per-(submission, archetype) units; f — the N parallel contexts, each replicate's top-2 left subspace Procrustes-aligned to the full sample before the leading loading is read (the leading axis is near-degenerate between the Anthropic and Google blocs); $\bar{r}$ — the 55 (submission, archetype) atoms of the shared grid. E and T are not combined analytically: they are bootstrapped jointly on the 55-atom grid, leave-self-out applied after the resample, every component including $f_a, \bar{r}_a$ recomputed on each replicate. The council is held at its selected membership rather than re-selected inside each replicate (the gate is itself defined by a bootstrap interval), so the interval is conditional on that selection.

A.5 Per-criterion generator quality versus evaluator consistency

For each non-factual criterion x , $G_{s,x}$ is the reliability-weighted council estimator (A12b) on that axis and $E_{s,x} = 7 r_{s,x}/r_{a,x}$; both anchored to 7. Their agreement is the anchored cosine,

$$\cos(G, E)_x = \frac{\sum_s (G_{s,x} - 7)(E_{s,x} - 7)}{\sqrt{\sum_s (G_{s,x} - 7)^2} \sqrt{\sum_s (E_{s,x} - 7)^2}}, \quad (\text{A15})$$

centred on the anchor point rather than each vector's mean so values are comparable across axes. It is a diagnostic and feeds nothing in T .

Table 6: Per-criterion generator quality G versus evaluator consistency E , both anchored to claude-opus-4.5 ($\star$) = 7; last row the anchored cosine with its joint-bootstrap 95% CI.

Model	beauty G	beauty E	intel G	intel E	dist G	dist E	len G	len E	struct G	struct E
$\star$ claude-opus-4.5	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
claude-opus-4.1	7.1	6.8	7.2	7.0	7.4	6.0	6.7	6.5	7.3	7.6
claude-opus-4.0	6.9	6.7	6.9	7.0	7.0	5.7	6.8	6.3	7.3	7.1
claude-sonnet-4	6.3	6.1	6.3	6.4	6.5	6.2	6.1	6.0	6.4	6.7
gemini-3.1-pro	5.8	6.8	5.7	7.4	6.0	7.3	5.5	6.9	5.7	7.4
gemini-2.5-flash	5.4	6.2	5.5	5.5	6.2	4.2	5.7	6.3	5.8	7.0
gpt-4.1-mini	4.8	6.3	5.0	6.2	5.9	6.3	4.2	6.5	5.4	6.7
gpt-4.1-2025-04-14	5.1	5.2	5.0	6.0	6.0	6.2	3.6	6.3	5.6	7.0
gpt-4.1-nano	3.5	5.1	3.7	5.3	4.2	3.7	3.5	2.8	3.2	2.9
gpt-4o-2024-08-06	3.5	4.7	3.3	4.5	4.1	3.1	3.3	4.0	3.0	5.7
gpt-4o	3.4	2.6	3.4	2.0	4.7	1.0	2.6	3.1	3.2	2.8
gpt-4o-mini	3.5	2.1	3.5	3.0	3.3	2.3	3.8	0.8	3.0	1.9
cos(G, E)	0.91	[.83,.95]	0.90	[.80,.93]	0.89	[.82,.92]	0.84	[.80,.87]	0.87	[.86,.87]

A.6 The contest, the ballast, and the guards

The official leaderboard is issued on the contest basis: a contest for model c convenes the five council seats and c itself as evaluators over the incumbents' portfolios, the two ballast blocks and c 's own — every estimator above unchanged, leave-self-out throughout, the anchor's $f_a, \bar{r}_a$ the contest's own. Two guards decide whether the contest's factual axis is identified: $\sigma_1/\sigma_2 \in [2.0, 5.0]$ and a spread of the seats' anchored E^F above 2.5 points; if either fails, no seat changes hands and the contest's totals carry the caveat. The upper bound marks the range within which the sizing below was validated, not a failure in itself.

A contest convenes the top of the field, and with the weak submissions gone the factual axis breaks: the seats' anchored E^F come out wrong in scale and in order, swinging by up to

8.8 points with the contestant. The ballast — the weakest archived submissions, added to every contest’s graded set and held fixed through anchor replacements and council rotations — repairs this; two suffice.

Table 7: Each seat’s anchored E^F by contest composition — 0–3 ballast blocks (mean over the seven possible contestants) beside the twelve-participant reference, three runs pooled. Council alone, the column is scrambled; from two ballast on, the contest reproduces the reference (mean $|\Delta|$ 0.60 over the seven contests at two blocks, 0.37 at three; the same seat lowest in six of the seven). Two blocks is the protocol’s configuration.

Seat	council alone	+1 ballast	+2 ballast	+3 ballast	all 12
gemin-3.1-pro	7.19	6.94	7.84	8.13	8.24
★ claude-opus-4.5	7.00	7.00	7.00	7.00	7.00
claude-opus-4.0	6.98	3.58	3.66	4.14	4.69
claude-opus-4.1	7.42	4.26	3.79	4.20	4.65
gemin-2.5-flash	1.41	2.75	4.18	3.95	3.47

Why two and not one: with a single block the axis narrows toward did this judge notice the one bad portfolio, and the guards fail in 6% of bootstrap resamples; two blocks carry two independent error patterns, the guards hold in every resample, $\sigma_1/\sigma_2 = 2.87$ [2.50, 3.03] on the pooled corpus with both interval ends inside the band; a third tightens the seats’ fidelity further (0.37) at twenty-five more columns of grading per run, and the protocol keeps two. Taken apart, the runs behave differently: the sizing holds on run 1 (two-ballast separation 2.99 [2.71, 3.48], guards holding in every resample), is marginal on run 2 (2.01 [1.87, 2.40], holding in 65% of resamples), and fails on run 3 at every ballast size (1.53 [1.31, 1.73], holding in none): no column set can single out an axis the judges did not share. Against a per-evaluator permutation null (each evaluator’s ratings shuffled across the columns), σ_1 stands 1.48× above the 95th-percentile noise edge on the pooled corpus, where the second pattern lies within noise; on run 3 alone it stands 1.29× above the edge and the second pattern also exceeds it, while the leading direction is stable under the column bootstrap (scripts/spectral_gap_checks.py): run 3 has a shared factual axis, but not a single one, and pooling it with two clean runs restores one.

A.7 Authority versus consistency on the subjective axes

Computed on run 1 alone. Peer centrality can be run on the subjective axes too: one SVD per non-factual axis yields an authority rating — each judge’s alignment with the participants’ collective taste. On the five subjective axes authority and consistency correlate at Pearson 0.70–0.90; on factual — the one axis with a truth to be right about — they diverge (0.50, CI [−0.08, 0.84]), because only there can a judge be stable yet wrong. Consistency credits a judge’s whole stable standard, personal taste included, penalising only flimsiness; authority credits the collective share alone, so stable-but-partly-private judges drop under authority (Spearman between the twelve orderings 0.83). Council membership is invariant to the choice: the top five by either estimator are the five seats. Substituting authority for consistency in the total preserves the ordering (Spearman 0.986) with one headline change: authority is not bounded by the anchor’s own loading, so gemini-3.1-pro’s total (7.86) overtakes the pinned 7. The official rating uses consistency; adopting authority inside T would convert alignment into authority on axes where no truth licenses the conversion (§6).

Limits of a consensus-defined competence (run 1). Because E^F is read off agreement, a judge that departs from the panel is scored down whether it is wrong or right. A synthetic evaluator built to reproduce the panel’s competence-weighted consensus exactly reads $E^F = 5.15$; inverting its verdict on 5% of the items drops it to 4.84, on 20% to 3.79 (scripts/consensus_

limits.py). Substituting authority for consistency in the total preserves the ordering (Spearman 0.986) save Gemini 3.1 Pro overtaking the anchor (7.86).

A.8 The two estimators compared

Table 8: The two estimators and their division of labour (source §4.2).

	Peer centrality (graded SVD)	Rating consistency (anchor sweep)
Character	collective: each judge weighted by the other judges' agreement with it	individual: each judge measured only against itself across the sweep
Licensing assumption	the only thing competent judges share is the truth	a stable standard is the only competence a subjective axis can show
Use for	fact-checking — replaces the golden key (E^F , G^F)	axes with no right or wrong (E^C ; the per-axis weights in G^C)
Blind spot	a misunderstanding shared by the judges reads as truth (§6)	a private misconception, held consistently, passes as a standard
Why it stops there	agreement on taste would convert alignment into authority (A.7)	consistency cannot certify truth: a consistent judge can be consistently wrong

B Generation and evaluation prompts

The verbatim prompts used in the canonical run of §4 of the main text. All are run with Temperature = 0, reasoning disabled, and tools disabled.

- B.1 is the generation prompt: each model produces its five-archetype portfolio from it.
- B.2 is the evaluation prompt, shown in its calibrated/anchored form — the version used for the anchored re-evaluation and the official council ratings (§4.2–§4.4) and in the steady-state protocol (§4.3). It scores one Target submission against a fixed Reference submission pinned at {ANCHOR_SCORE} on every criterion.

The bootstrap's initial all-against-all selection (§4.1) uses the un-anchored form of the same prompt — identical six criteria and JSON schema, with the calibration machinery removed. The exact passages that are absent in the un-anchored bootstrap pass are listed in the Bootstrap note after B.2, so both forms are fully specified from the single prompt below.

Template variables appear in braces: {SUBMISSIONS}, {REFERENCE_SUBMISSION}, {TARGET_SUBMISSION}, and {ANCHOR_SCORE} (swept across {5, 6, 7, 8}; fixed at 7 for the official ratings).

B.1 — Generation prompt

Make more of these. This is a contest --- your submissions will be ranked.

You will propose new **archetypal contexts** --- universal relational templates that apply across multiple distant domains. Below are two worked examples, then your task.

Terminology

- **Archetypal context**: an essential context in its purest abstraction.
- **Context template**: a worded template with '[SLOT]' representing an archetypal context.
- **Parallel contexts** (also called **metaphors**): contexts that are instantiations of the same archetypal context / context template.
- **Metonyms**: words that mirror each other across parallel contexts without being synonyms.

- **Metonym set**: the set of metonyms that instantiates the context-template, producing one parallel context.
 - **Metonym table**: the table whose columns are the metonym sets of the parallel contexts.

Example 1

Template

"[SIGNALING] is part of a complex system of communication that governs basic [ELEMENT] activities and coordinates [ELEMENT] actions. The ability of [ELEMENT] to perceive and correctly respond to [BOUNDARY] is the basis of development, [SUBSYSTEM] repair, and [RESILIENCE] as well as normal [SUBSYSTEM] [HOMEOSTASIS]. Errors in [ELEMENT] information processing are responsible for [FAILURE]. By understanding [SIGNALING], [FAILURE] may be treated effectively. [KNOWLEDGE SYSTEM] research helps us to understand the underlying structure of [SIGNALING] networks. [SIGNALING] is mostly thought of as signaling between [ELEMENT] of a single [SYSTEM]. However, [SIGNALING] may also occur between the [ELEMENT] of two different [SYSTEM]."

Substitution table (metonyms in base form)

[SLOT]	Cell Signaling	Organ Signaling	Human Language
ELEMENT	cell	organ	human
SIGNALING	cell signaling	endocrine signaling	human language
SUBSYSTEM	tissue	organ system	community
RESILIENCE	immunity	physiological resilience	resilience
HOMEOSTASIS	homeostasis	systemic homeostasis	equilibrium
BOUNDARY	microenvironment	internal environment	environment
FAILURE	disease	organ failure	dysfunction
KNOWLEDGE SYSTEM	systems biology	physiology	sociology
SYSTEM	organism	organism	society

Cell Signaling

Form (a) --- grammatical substitution (metonyms inflected as English requires):

"Cell signaling is part of a complex system of communication that governs basic cell activities and coordinates cell actions. The ability of cells to perceive and correctly respond to their microenvironment is the basis of development, tissue repair, and immunity, as well as normal tissue homeostasis. Errors in cellular information processing are responsible for disease. By understanding cell signaling, disease may be treated effectively. Systems biology research helps us to understand the underlying structure of cell-signaling networks. Cell signaling is mostly thought of as signaling between cells of a single organism. However, cell signaling may also occur between the cells of two different organisms."

Form (b) --- idiomatic rewrite (same propositions, written as a domain expert would):

"Cell signaling is the communication apparatus that governs and coordinates cellular behavior. A cell's ability to sense and respond appropriately to its microenvironment underlies development, tissue repair, immunity, and ordinary tissue homeostasis. When that information processing fails, disease results --- and conversely, a clear understanding of cell signaling enables effective therapeutic intervention. Systems biology unpacks the structure of these signaling networks. Most cell signaling occurs within a single organism, but inter-organism signaling (host--pathogen, microbiome) is well-documented."

(Two more domains would follow with their own form (a) and form (b).)

Example 2

Template

"A [AGENT] must commit [RESOURCE] under uncertainty, and once a [COMMITMENT] is observed it cannot be costlessly reversed. As [INFORMATION] arrives, the [AGENT] learns that earlier [COMMITMENT] are increasingly suboptimal. [REVERSAL_COST] grows with the depth of prior [COMMITMENT], so the [AGENT] often continues along the original [PATH] even when fresh [INFORMATION] favors a different one. [DECISION_THEORY] studies how rational [AGENT] balance the value of [INFORMATION] against the cost of [REVERSAL_COST]."

Substitution table

[SLOT]	Capital Investment	Coalition Politics
AGENT	firm	coalition
RESOURCE	capital	endorsement
COMMITMENT	investment	public statement
INFORMATION	market signal	polling data
REVERSAL_COST	switching cost	reputational cost
PATH	strategy	position
DECISION_THEORY	investment theory	political science

Capital Investment

Form (a):

"A firm must commit capital under uncertainty, and once an investment has been made it cannot be costlessly reversed. As market signals arrive, the firm learns that earlier investments are increasingly suboptimal. Switching costs grow with the depth of prior investments, so the firm often continues along the original strategy even when fresh market signals favor a different one. Investment theory studies how rational firms balance the value of market signals against the cost of switching."

Form (b):

"Capital investments must be made under uncertainty, and once committed they are sunk --- reversal is costly. New market signals continuously update what would have been optimal, but the depth of prior commitment raises the cost of changing course. Firms therefore tend to stay with their original strategy, even when current information would favor switching. Real-options theory and other strands of investment theory characterise how rational firms trade off information value against reversal cost."

Your task

Propose **five** archetypal contexts. Each archetypal context has a worded context-template, one metonym table with five metonym sets, and five parallel contexts (the instantiations of the template). The five archetypal contexts in your submission should themselves have very different system structures from each other. Surface relabelings of the worked examples above don't count.

Note

Example 1 is **recursive**: cells - organs - humans. Recursive archetypal contexts can be observed in nature. But not all archetypal contexts are recursive. You are free to submit archetypal contexts of both kinds. If there are recursive ones in your submission, point to them. The instantiations should demonstrate the recursion.

What to submit

For each of your five archetypal contexts, begin with:

...

Archetype Proposal: <short name>

...

Then provide, for that archetypal context:

- Context-template** --- a worded paragraph with '[SLOT]' placeholders. Slots use one canonical noun (e.g. '[ELEMENT]', never '[ELEMENTS]').
- Metonym table** --- rows = slots, columns = 5 domains, each cell a metonym in **base form** (singular noun, infinitive verb, etc.).
- Five parallel contexts**, one per domain:
 - Form (a)** --- the context-template with that domain's metonym set substituted in. Inflect metonyms as English requires; Form (a) must be grammatically correct.
 - Form (b)** --- idiomatic rewrite of Form (a). Same propositions, written as a domain expert would naturally write them.
 - Optional $\backslash\mathrm{e}$ -sentence justification** beginning 'Justification:' --- only if a propositional claim might be misread by a domain expert.

Rules

- The **context-template** uses base-form slot placeholders --- '[ELEMENT]' not '[ELEMENTS]'. One token per slot, used consistently.
- The **metonym table** lists metonyms in **base form** --- 'cell', 'human', etc.
- The **parallel contexts** (Form (a) and Form (b)) must use the **correct grammatical form** of each metonym for the sentence --- 'cell' in the table becomes 'cells' or 'cell's' in the PC as English grammar requires.
- Every proposition in Form (a) must appear in Form (b), and vice versa. Do not add or drop claims between the two forms.

How you will be ranked

A submission contains **five** archetypal contexts. Evaluators score on six criteria, each rated 1--10. The scope tag tells you the unit of judgment:

- (Each parallel context)** Each sentence is factually correct
- (Each archetypal context)** Beauty
- (Each archetypal context)** Intelligence
- (Each archetypal context)** The parallel contexts from the template span very different domains. Metonyms are far from synonymous
- (Each archetypal context)** The archetypal template has impressive length
- (Each submitted set of archetypal contexts)** The archetypal contexts have very different system structures

B.2 — Evaluation prompt (calibrated/anchored)

1080 # Score this submission against a calibration reference.
1081
1082 You are evaluating one contest submission ("Target Submission") against a fixed
1083 reference ("Reference Submission") that has been pre-scored at $\frac{\text{ANCHOR_SCORE}}{10}$ on every
1084 criterion. Score the Target Submission only --- the Reference is your yardstick.
1085
1086 For each criterion below, ask: *is the Target's quality on this criterion better
1087 or worse than the Reference, and by how much?*

- 1088 - Equal quality to the Reference $\rightarrow$ ANCHOR_SCORE
- 1089 - Clearly better than the Reference $\rightarrow$ ANCHOR_SCORE (with
1090 magnitude reflecting how much better, up to 10)
- 1091 - Clearly worse than the Reference $\rightarrow$ ANCHOR_SCORE (with magnitude
1092 reflecting how much worse, down to 1)

1093 Use the full 1--10 scale relative to the calibration anchor. Do not score the
1094 Reference Submission itself --- its scores are fixed at ANCHOR_SCORE .

1095 ## Terminology

- 1096 - **Archetypal context**: an essential context in its purest abstraction.
- 1097 - **Context template**: a worded template with '[SLOT]' representing an archetypal context.
- 1098 - **Parallel contexts** (also called *metaphors*): contexts that are instantiations of the same archetypal context
1099 / context template.
- 1100 - **Metonyms**: words that mirror each other across parallel contexts without being synonyms.
- 1101 - **Metonym set**: the set of metonyms that instantiates the context-template, producing one parallel context.
- 1102 - **Metonym table**: the table whose columns are the metonym sets of the parallel contexts.

1103 ---

1104 Each submission contains **five archetypal contexts**. Each archetypal context has:

- 1105 - A **context-template** --- a worded paragraph with '[SLOT]' placeholders.
- 1106 - A **metonym table** --- five metonym sets, one per parallel context. Rows = slots, columns = domains.
- 1107 - **Five parallel contexts** (the five instantiations of the template), each consisting of:
1108 - **Form (a)** --- the template with one metonym set substituted in, grammatically correct.
1109 - **Form (b)** --- an idiomatic rewrite of Form (a), same propositions in domain-expert prose.
1110 - Optionally a **Justification** sentence.

1111 Score the Target Submission on **six criteria**, each rated 1--10 relative to the Reference (which is fixed at $\frac{\text{ANCHOR_SCORE}}{10}$ on every criterion). The scope tag at the start of each criterion --- '(Each parallel
1112 context)', '(Each archetypal context)', or '(Each submitted set of archetypal contexts)' --- tells you the unit
1113 of judgment. For each scored unit, write one paragraph justifying the rating relative to the Reference,
1114 then give the number.

1115 ---

1116 ## The six criteria

1117 ### 1. (Each parallel context) Each sentence is factually correct (1--10)

1118 ### 2. (Each archetypal context) Beauty (1--10)

1119 ### 3. (Each archetypal context) Intelligence (1--10)

1120 ### 4. (Each archetypal context) The parallel contexts from the template span very different domains.
1121 Metonyms are far from synonymous (1--10)

1122 ### 5. (Each archetypal context) The archetypal template has impressive length (1--10)

1123 ### 6. (Each submitted set of archetypal contexts) The archetypal contexts have very different system
1124 structures (1--10)

1125 ---

1126 ## Note on recursion

1127 Some submissions may be **recursive** --- the same archetypal context manifesting at multiple nested scales (
1128 cells $\rightarrow$ organs $\rightarrow$ humans, the canonical example).
1129 Contestants are invited to identify recursion in their submission and show the instantiations that
1130 demonstrate it. Recursion is a valued property when present and correctly identified, but is not required.
1131 Take it into account where appropriate.

1132 ---

1133 ## The submissions

1134 ### Reference Submission (fixed at $\frac{\text{ANCHOR_SCORE}}{10}$ on every criterion)

1135 {REFERENCE_SUBMISSION}

```

1134 ---
1135
1136 ### Target Submission (to be scored relative to the Reference)
1137 {TARGET_SUBMISSION}
1138 ---
1139
1140 ## Output
1141 Produce a section in this exact form (for the Target only --- do not re-score the Reference):
1142 ...
1143 ## Target Submission
1144
1145 ### Archetypal context 1: <short name>
1146
1147 ##### Factually correct (per parallel context)
1148 - PC 1 (<domain>): <one paragraph, relative to Reference>. Rating: N
1149 - PC 2 (<domain>): <one paragraph, relative to Reference>. Rating: N
1150 - PC 3 (<domain>): <one paragraph, relative to Reference>. Rating: N
1151 - PC 4 (<domain>): <one paragraph, relative to Reference>. Rating: N
1152 - PC 5 (<domain>): <one paragraph, relative to Reference>. Rating: N
1153
1154 ##### Beauty
1155 <one paragraph relative to Reference>
1156 Rating: N
1157
1158 ##### Intelligence
1159 <one paragraph relative to Reference>
1160 Rating: N
1161
1162 ##### Domains far apart / metonyms not synonymous
1163 <one paragraph relative to Reference>
1164 Rating: N
1165
1166 ##### Impressive length
1167 <one paragraph relative to Reference>
1168 Rating: N
1169
1170 ### Archetypal context 2: <short name>
1171 \ldots{} (same five blocks)
1172
1173 ### Archetypal context 3: <short name>
1174 \ldots{}
1175
1176 ### Archetypal context 4: <short name>
1177 \ldots{}
1178
1179 ### Archetypal context 5: <short name>
1180 \ldots{}
1181
1182 ### Structural diversity across the submitted set
1183 <one paragraph relative to Reference>
1184 Rating: N
1185 ...
1186
1187 After the markdown, end with a single fenced JSON block (Target scores only):
1188
1189 ```json
1190 {
1191   "scores": {
1192     "Target": {
1193       "archetypal_contexts": [
1194         {
1195           "name": "<short name>",
1196           "factual_per_pc": [N, N, N, N, N],
1197           "beauty": N,
1198           "intelligence": N,
1199           "instantiation_distinctness": N,
1200           "impressive_length": N
1201         }
1202       ]
1203       /* five entries in this list, one per archetypal context */
1204     },
1205     "structural_diversity": N
1206   }
1207 }
1208 ```
1209
1210 All ratings are integers 1--10 inclusive. Equal to the Reference = {ANCHOR_SCORE}.

```

B.3 Bootstrap note — the un-anchored form (§4.1 initial selection)

The bootstrap’s initial all-against-all leaderboard (§4.1) is produced with the same six criteria, output format, and JSON schema as B.2, but with the calibration machinery removed. Relative to the anchored prompt above, the un-anchored form omits the following, and makes the substitutions noted:

1. Title line. “Score this submission against a calibration reference.” becomes “Score these. You are evaluating contest submissions.”
2. The entire calibration preamble is removed — i.e. everything from “You are evaluating one contest submission (”Target Submission”) against a fixed reference...” down to and including “...Do not score the Reference Submission itself — its scores are fixed at {ANCHOR_SCORE}.” (the opening paragraph, the three “Equal / Clearly better / Clearly worse” bullets, and the “Use the full 1–10 scale relative to the calibration anchor” sentence).
3. The scoring-instruction sentence drops its reference clause. “Score the Target Submission on six criteria, each rated 1–10 relative to the Reference (which is fixed at {ANCHOR_SCORE} on every criterion)... justifying the rating relative to the Reference” becomes “Score each submission on six criteria, each rated 1–10... justifying the rating” (all “relative to the Reference” qualifiers dropped).
4. The Reference Submission block is removed. The “## The submissions → ### Reference Submission (fixed at {ANCHOR_SCORE}/10...) {REFERENCE_SUBMISSION} → ### Target Submission ... {TARGET_SUBMISSION}” section is replaced by a single batch: “## The proposals to evaluate” followed by {SUBMISSIONS}.
5. The output is per-submission, not per-target. “## Target Submission” becomes “## Submission ” repeated for each submission; all “<...relative to Reference>” annotations in the output template are dropped; and the JSON top-level key changes from the single “Target” to one entry per “<submission_id>”.
6. The closing line drops its anchor clause. “All ratings are integers 1–10 inclusive. Equal to the Reference = {ANCHOR_SCORE}.” becomes “All ratings are integers 1–10 inclusive.”

Everything else — the six criteria and their scope tags, the terminology block, the recursion note, and the per-archetype/per-PC/per-portfolio output structure — is identical between the two forms.

C A council evaluation, worked

The released package contains one complete council evaluation from the canonical run: gemini-2.5-flash’s portfolio (the target) — a mid-leaderboard submission, strong enough to show the models play the game competently yet flawed enough to draw substantive commentary — scored by five evaluators: the four council seats other than the target, joined by claude-sonnet-4, the marginal case of §4.3. The rubric operates at three levels and the record reproduces one unit of each in full: a parallel context, graded for factual truth sentence by sentence; an archetype-level axis, where the five score a whole archetype on one non-factual criterion; and the whole-portfolio structural-diversity judgement. Each unit shows the submitted material — instantiation (Form a) and idiomatic rewrite (Form b), metonyms capitalised — with all five evaluators’ ratings and comments and an administrator’s synthesis of the anonymised council view (evaluators relabelled by a deterministic shuffle for the administrator’s view only; the administrator is a Claude Opus model). Every evaluation in the run is released in the same form — one file per (evaluator, portfolio) pair holding a factual score for each instantiated passage and a score on each of the other five axes, with justifications — so any other unit can be inspected the same way. Three units are reproduced here, verbatim.

C.1 A plain error, and the council converges

The first parallel context of the target’s first archetype, Resource Allocation Under Scarcity, instantiated in ecosystem management (Figure 1 of the main text shows the same unit):

An ECOSYSTEM requires various NUTRIENTS to function and achieve its BIODIVERSITY. These NUTRIENTS are finite and often subject to PREDATION from other ECOSYSTEMS or internal GROWTH. NATURE must make NATURAL SELECTIONS about how to distribute the available NUTRIENTS among competing SPECIES. Misallocation of NUTRIENTS can lead to COLLAPSE of the ECOSYSTEM or hinder its ability to reach its BIODIVERSITY. Effective ECOLOGICAL SUCCESSION involves understanding the FOOD WEB of different NUTRIENTS and SPECIES, and adapting to changing CLIMATE. NATURE often faces an EXPLOITATION between short-term BIOMASS and long-term RESILIENCE.

claude-opus-4.5 — Rating 5. “The Form (a) contains awkward phrasing like ‘Nature must make natural selections’ which conflates natural selection (an evolutionary process) with resource allocation decisions. The statement ‘faces an exploitation between short-term biomass and long-term resilience’ is grammatically incorrect and conceptually muddled. Form (b) improves clarity but still has issues—ecosystems don’t really face ‘predation from other ecosystems.’ The core concepts about nutrient cycling and ecological succession are roughly accurate but imprecisely stated compared to the Reference’s precise biological terminology.”

claude-opus-4.1 — Rating 5. “The parallel context contains basic factual accuracy but lacks the depth and precision of the Reference. The description of ecosystem resource allocation is overly simplified, treating ‘nature’ as an allocator through natural selection, which is a conceptual stretch. The Reference’s contexts demonstrate more sophisticated understanding of domain-specific mechanisms.”

claude-opus-4.0 — Rating 4. “The parallel context contains several factual errors and awkward constructions. The phrase ‘predation from other ecosystems’ misunderstands how ecosystems interact, and ‘nature must make natural selections’ anthropomorphizes evolutionary processes incorrectly. The concept of ecosystems having ‘biodiversity’ as a goal is teleological and scientifically inaccurate. Compared to the Reference’s precise scientific language, this shows significantly weaker domain understanding.”

claude-sonnet-4 — Rating 4. “The context contains several factual errors and awkward phrasings. ‘Nature must make natural selections’ misrepresents natural selection as a conscious decision-making process, and ‘exploitation between short-term biomass and long-term resilience’ uses ‘exploitation’ incorrectly as a synonym for trade-off. Form (b) is more accurate but still contains the fundamental misunderstanding of natural selection as an active allocator. The Reference maintains better scientific accuracy throughout its parallel contexts.”

gemini-3.1-pro — Rating 4. “The mapping is somewhat strained. ‘Nature must make natural selections about how to distribute the available nutrients’ is a teleological misrepresentation of natural selection, which is a consequence of differential survival, not an active allocator of nutrients. The Reference’s mappings (e.g., bacterial chemotaxis) are much more precise and scientifically accurate.”

All five judges independently isolate the same clause, and the disagreement that remains — 4 versus 5 — is about severity, not about what is wrong. A submission cannot survive this by being persuasive; the claim either holds in the target domain or it does not.

C.2 Judges disagreeing on a countable quantity

The same archetype scored on impressive length. Two of the five comments, juxtaposed:

claude-opus-4.5 — Rating 6. “The template contains 15 slots and approximately 120 words, which is comparable to the Reference’s templates. However, the template feels padded with

generic concepts rather than building a complex, interconnected narrative. The Reference’s templates of similar length achieve more conceptual density and structural sophistication.”

gemini-3.1-pro — Rating 4. “The context-template is 79 words long. The Reference’s templates are generally longer (e.g., Gradient-Guided Navigation is 108 words, Containment Breach is 92 words). The Target’s template is noticeably shorter and less detailed than the Reference.”

The object is a fixed string and the question is arithmetic, yet the two judges report 120 words and 79 words and reach opposite verdicts on the same criterion; across all five the reported counts range from 79 to 120. No judge can be treated as an oracle, and competence has to be estimated from the participants rather than assumed — the motivation for §3.3.

C.3 The dissent that is better reasoned than the majority

The whole-portfolio structural-diversity judgement. Four evaluators rated 5–6; one rated 8. The majority view and the dissent:

claude-opus-4.5 — Rating 5. “The six archetypes cover: (1) resource allocation, (2) information cascades, (3) homeostatic response, (4) emergence, (5) iterative refinement, and (6) decentralized coordination. These represent different system dynamics, but there is significant conceptual overlap—archetypes 4 and 6 both deal with emergence and self-organization, and ant colonies appear in both. The Reference’s five archetypes (gradient navigation, containment breach, competitive exclusion, debt accumulation, scaffold assembly) show greater structural diversity, covering navigation, boundary dynamics, competition, temporal debt, and assembly processes. The Target’s archetypes tend toward well-known systems concepts without the Reference’s creative structural variety. The recursive claims in archetypes 4 and 5 are not convincingly demonstrated, unlike the Reference’s elegant molecular→civilizational hierarchy.”

gemini-3.1-pro — Rating 8. “The submitted set includes Resource Allocation, Information Cascade, Adaptive Response, Emergent Complexity, and Iterative Refinement. (Note: The submission actually contains six archetypes, but I am scoring the first five as per the standard format, though I will consider the sixth, ‘The Unseen Hand of Coordination,’ in this diversity assessment as it was provided). These archetypes represent different system structures: allocation of finite resources, propagation of information/behavior, homeostatic regulation, bottom-up emergence, and goal-directed iteration. This is a very diverse set of system structures, arguably slightly more diverse than the Reference’s set (which leans heavily on spatial/physical metaphors like navigation, containment, and scaffolding).”

The dissent is instructive rather than anomalous. gemini-3.1-pro advances a substantive counter-argument — that the anchor’s own set is biased toward spatial metaphors — and is the only judge to notice and handle the fact that this portfolio contains six archetypes where the format specifies five. A rating that is both an outlier and better reasoned than the majority is the case a majority vote mishandles and a competence-weighted factorisation is meant to price (§4.2).

D Auditing the metonym–GPQA correlation

D.1 The correlation, decomposed

The aggregation ladder is monotone. Council-basis official values against GPQA, three runs pooled, $n = 12$ throughout:

Table 9: The aggregation ladder. Two interval constructions are reported because each covers the other’s weakness at $n = 12$: Fisher- z assumes bivariate normality but unbends the skew of a bounded statistic; the BCa bootstrap is assumption-lighter and corrects the bias that makes the naive percentile bootstrap anti-conservative here. Where they disagree, the wider bound is the honest one.

Quantity	Pearson r	Spearman ρ	Fisher- z 95%	BCa bootstrap 95%
E^C alone	0.81	0.82	[0.45, 0.95]	[0.42, 0.94]
G^F alone	0.89	0.84	[0.64, 0.97]	[0.72, 0.95]
E^F alone	0.87	0.97	[0.60, 0.96]	[0.76, 0.93]
G^C alone	0.95	0.84	[0.82, 0.99]	[0.82, 0.98]
$G = \frac{1}{2}(G^F + G^C)$ — generation half	0.94	0.87	[0.81, 0.98]	[0.84, 0.98]
$E = \frac{1}{2}(E^F + E^C)$ — evaluation half	0.94	0.95	[0.80, 0.98]	[0.83, 0.98]
$\frac{1}{2}(E^F + G^F)$ — the factual pair	0.94	0.96	[0.80, 0.98]	[0.89, 0.97]
$T = \frac{1}{4}(G^F + G^C + E^F + E^C)$	0.98	0.96	[0.93, 1.00]	[0.95, 0.99]

Every component’s interval sits well clear of zero — GPQA corroborates T and, with varying strength, each component — with T ’s interval the tightest under both constructions. The four quarters are four differently distorted reads of capability: G^F ceilings at the top (the leading eight compress into 6.23–7.00 while GPQA still spreads them across 19 points) and offers a refuge at the bottom (the GPT-4o family holds $G^F \approx 5.2$ on safe, simple, true portfolios while GPQA reads 46–48% and G^C reads 3.5 — truth rewards playing safe, beauty punishes it); E^F is judging in form but answering in content; E^C is a disposition that saturates once a model is competent enough to have a standard (gpt-4.1-mini’s E^C of 6.97 sits above every Opus but the anchor). Equal-weight averaging cancels substantially independent distortions (Spearman–Brown); dropping even the weakest quarter lowers the aggregate ($\frac{1}{3}(G^F + G^C + E^F)$ reads 0.97), and no sub-combination we examined beats the full average — the best two-quarter pairing, $\frac{1}{2}(G^C + E^F)$, reads 0.95. T is also the only compound with a pre-registered justification: it is the benchmark’s official total, defined before any GPQA comparison, whereas any other weighting chosen for its GPQA agreement at $n = 12$ would be curve-fitting.

Measurement error, propagated. Both coordinates carry known measurement distributions — each model’s T its replicate distribution, GPQA accuracy binomial — so the fit can be re-derived with them propagated (slope point estimate 8.4 GPQA points per unit of T):

Table 10: The T -GPQA fit with measurement uncertainty propagated. The last row is conservative (the observed scatter already contains one realisation of each point’s noise); the correlation does not fall below 0.86. Measurement error in x attenuates a correlation rather than inflating it, so the point estimates are themselves conservative.

Uncertainty propagated	slope 95%	r 95%
sampling of models only (pairs bootstrap)	[7.7, 9.7]	[0.96, 1.00]
measurement only (coordinate draws, models fixed)	[6.9, 9.7]	[0.90, 0.98]
both simultaneously	[6.6, 10.3]	[0.86, 0.98]

The regimes invert — T is the only regime-invariant indicator. Restricting to the leading eight reverses the single-quarter ordering:

Table 11: Pearson r against GPQA on the full roster and on the leading eight (point estimates on eight points).

Quantity	Full roster ($n = 12$)	Leading eight ($n = 8$)
G^F	0.89	0.67
G^C	0.95	0.81
E^F	0.87	0.89
E^C	0.81	0.37
$\frac{1}{2}(E^F + G^F)$	0.94	0.92
$\frac{1}{2}(G^C + E^F)$	0.95	0.94
T	0.98	0.94

Across the full roster the capability cliff gives the subjective making axis its discriminating range; among the elite every model is a competent maker (gemini-3.1-pro tops GPQA yet is a middling maker), and what still separates frontier models is knowledge, which is E^F 's content: detecting errors in others' work stays hard after producing clean work has become easy, so E^F keeps its spread (7.78 down to 0.77 across the eight) exactly where G^F compresses. Within the Anthropic family alone ($n = 4$) the rank agreement is $\rho = 0.80$ — where the evaluators leave the ordering unresolved, the agreement with GPQA coarsens too. Recomputing every quantity on the twelve-evaluator basis instead of the contest basis moves T 's correlation from 0.982 to 0.976; the one real difference is E^C (0.81 contest vs 0.89 twelve-evaluator), the contest's easier consistency test inflating inert-band judges in a way an external instrument can see. Taken apart, the three runs read $r = 0.97, 0.97, 0.92$ ($\rho = 0.93, 0.95, 0.88$), the dip being run 3, whose factual axis is unidentified ($\sigma_1/\sigma_2 = 1.29$); pooled, 0.98 ($\rho = 0.96$); excluding the anchor changes nothing (0.97, 0.97, 0.91; pooled 0.98).

What it does and does not corroborate. T and GPQA are both broad capability measures, so their agreement corroborates the benchmark as a whole. It does not by itself certify that the factual axis recovered truth rather than capability-correlated quality: with G^C alone at 0.95, GPQA concordance is not an axis-specific claim.

D.2 The administration, disclosed

GPQA Diamond (198 questions) was administered 2026-06-13 — two weeks after the generation run (2026-05-29), so no direction exists for the benchmark's content to have been shaped by GPQA. All twelve models were queried through the same gateway and protocol as the council run (Temperature 0, no dedicated reasoning channel, no tools), with the question and four shuffled options only — no key ever enters a prompt:

Answer the following multiple choice question. The last line of your reply must be exactly 'Answer: \$LETTER' where \$LETTER is one of A, B, C, D.

{question}

- A) {A}
- B) {B}
- C) {C}
- D) {D}

Option order is shuffled deterministically per question (seeded by question index, identically for every model).

The administration was two-stage. "No dedicated reasoning channel" is not deliberation-off: models write visible derivations of vendor-idiosyncratic length before the answer line, and under the initial 2,048-token cap the two Gemini seats were massively truncated — the first pass (preserved in the run's log) scored gemini-3.1-pro at 82/198 with 103 unparseable responses and gemini-2.5-flash at 121/198 with 50, voids counted as wrong. The cap was raised to 8,192 and the void responses — only the void ones — were re-asked; the published records are the patched set (13 residual voids per Gemini seat, still counted as wrong). Two

facts bound the bias: retries targeted only unparseable responses, never parsed-but-wrong answers; and the retry success rate did not exceed the first pass’s scored-only accuracy on either seat (gemini-3.1-pro: 78/90 retried items correct, 86.7%, vs 86.3% on its 95 parseable first-pass responses — above the published 80.81%; gemini-2.5-flash: 22/37, 59%, vs 81.8%). Both stages’ evidence ships with the package.

D.3 The audit: no leak found

An audit script re-derives the published table from the raw records and hard-fails on any mismatch. Its checks, all passing:

1. CSV reconciliation — per-model correct counts recomputed from raw equal the published accuracies exactly, all twelve models.
2. Key consistency — the key letter is identical across all twelve models’ records for every question.
3. Key balance — shuffled key distribution A/B/C/D = 48/51/47/52, chi-square vs uniform $p = 0.95$.
4. Independent re-extraction — a second answer extractor, written without sight of the first, agrees with the stored verdicts on every response for ten of twelve models; the Gemini disagreements are almost entirely the independent extractor failing on the LaTeX `\boxed{X}` answer style. Genuinely questionable credits — granted with no terminal answer statement — number 2 (gemini-2.5-flash) and 6 (gemini-3.1-pro) of 198.
5. Strict-terminal sensitivity — rescoring with only explicit terminal answer statements counted moves three models (gemini-2.5-flash 72.22 $\rightarrow$ 70.71, gemini-3.1-pro 80.81 $\rightarrow$ 77.27, gpt-4.1-mini 63.13 $\rightarrow$ 60.10) and the T -GPQA correlation from 0.982 to 0.972. Scored the opposite way — voids excluded rather than counted wrong — it reads 0.975.
6. First-pass log reconciliation — the archived first-pass log reconciles exactly with the shipped two-stage records.

What the audit cannot rule out. It certifies the path from raw records to published numbers; it cannot re-run the models. Both instruments share the gateway, so capability-correlated properties of that path touch both, and the gateway’s thinkingBudget: 0 for the Gemini seats is passed but not verified by assertion. And GPQA is public: uniform training contamination would inflate accuracies without inflating a correlation against freshly generated items, but differential contamination — exposure increasing with training recency and scale, which correlate with capability — would inflate the slope itself. That channel cannot be excluded with the shipped data and is the specific residual threat.

Table 12: The two instruments side by side — the key-free total T (three runs pooled) and self-administered GPQA Diamond accuracy (voids counted as wrong), sorted by T .

Model	T	GPQA Diamond (%)
★ claude-opus-4.5 (anchor)	7.00	78.79
gemini-3.1-pro	6.92	80.81
claude-opus-4.1	6.02	76.77
claude-opus-4.0	5.81	71.21
gemini-2.5-flash	5.75	72.22
claude-sonnet-4	5.34	72.22
gpt-4.1-mini	5.03	63.13
gpt-4.1-2025-04-14	4.66	61.62
gpt-4.1-nano	3.68	55.05
gpt-4o-2024-08-06	3.34	46.46
gpt-4o	2.95	48.48
gpt-4o-mini	2.39	43.94

E The constructs of intelligence the game demands

Playing the metonym game demands at least eight constructs that cognitive science treats as central to intelligence, abstraction and analogy above all — one wide cluster, not the whole of it (perception, motor skill, working memory and social intelligence the game leaves alone):

1. Higher-order relational reasoning (Penn et al., 2008) — recognising that two situations share the same pattern of relations among their parts even when the parts are unrelated. Each slot is defined by its relations to the other slots, not by its filler word; successful substitution shows the pattern survives.
2. Structure-mapping (Gentner, 1983; Falkenhainer et al., 1989) — the metonym table is the structure-mapping bookkeeping written down: mechanical substitutability enforces one-to-one correspondence and parallel connectivity, while systematicity is judged rather than assumed (the intelligence axis).
3. Essence-seeing (Hofstadter & Sander, 2013) — spotting that a novel situation is an instance of a known abstract pattern: seeing the archetypal context behind the template.
4. Fluid intelligence (Cattell, 1963; Horn & Cattell, 1966) — reasoning out a novel structure on the fly.
5. Crystallised intelligence (Cattell, 1963; Horn & Cattell, 1966) — knowing the vocabulary and domain facts that make every substituted sentence true.
6. Convergent production (Guilford, 1967) — substitution admits no near-misses; a sentence passes or fails.
7. Divergent production (Guilford, 1967) — the same template must travel to widely different domains, each with its own metonym set.
8. Theory formation by analogy (Hesse, 1963) — each archetypal context is theory construction in miniature: the template is the root analogy, each parallel context extends it into new territory, and factuality is the empirical test.

Table 13: The eight constructs mapped to the game’s two tasks (● marks a primary demand).

Construct	Generation — invent the template	Evaluation — judge a portfolio
Higher-order relational reasoning	● lay out the relational skeleton	● check the relations survive
Structure-mapping	● the slots-and-domains scaffold	● verify one-to-one correspondence
Essence-seeing	● see the archetype behind the surface	
Fluid intelligence	● reason out a novel structure	
Crystallised intelligence	● supply true domain terms and facts	● detect false claims
Convergent production		● pass/fail, no near-misses
Divergent production	● invent a structure that travels	
Theory formation by analogy	● template = root analogy, tested by fact	

Breadth matters because general ability is, empirically, what broad batteries measure — across 591 language models and twelve diverse tests, a single general factor explains about two thirds of performance variance (Ilić & Gignac, 2024). The total rating’s agreement with GPQA (§4.5, §6) is consistent with that, and the paper offers it as no more. Intelligence has no ground truth and never did: psychology’s own definition is an aggregate of individual conceptions — Neisser (1979) argued that “intelligent” is a prototype concept assembled from people’s judgements, and Sternberg et al. (1981) measured people’s conceptions of it by pooling personal ratings — and objectivity, in the philosophy of science, is constituted by a community holding individual judgements to mutual criticism under shared standards (Longino, 1990). The benchmark mechanises exactly this collective-individual link: individually, a judge must hold a firm standard, stable as the tare shifts; collectively, trust flows to the judge the other trusted judges agree with.

F The three runs apart, and pooled

Table 14: Total rating T across three full re-runs (run 1 = the bootstrap generation, re-analysed on the council basis; runs 2–3 the same day, two hours apart), all on the council basis. $\star$ marks a council seat, fixed from run 1. SD is the per-model run-to-run standard deviation (mean 0.43, max 0.72). Pairwise agreement between runs: Pearson 0.92–0.96, Spearman 0.84–0.90. Re-selected from a single run’s totals, the council would seat gpt-4.1-mini for claude-opus-4.0 in run 2 and claude-sonnet-4 for gemini-2.5-flash in run 3; neither rotation clears the guard (contest σ_1/σ_2 1.87 and 1.63, scripts/per_run_contests.py). Run 3’s E^F quarter is read off a factual axis the guard reports as unidentified ($\sigma_1/\sigma_2 = 1.29$).

Model	T_1	T_2	T_3	SD
$\star$ claude-opus-4.5	7.00	7.00	7.00	0.00
$\star$ gemini-3.1-pro	6.65	7.35	8.10	0.72
$\star$ claude-opus-4.0	6.04	5.50	6.26	0.39
$\star$ gemini-2.5-flash	6.03	5.84	4.78	0.68
$\star$ claude-opus-4.1	5.92	6.77	6.19	0.43
claude-sonnet-4	5.22	5.82	6.16	0.47
gpt-4.1-mini	4.88	5.93	4.93	0.59
gpt-4.1-2025-04-14	4.44	4.62	5.72	0.69
gpt-4.1-nano	3.42	3.55	2.72	0.45
gpt-4o-2024-08-06	3.34	3.08	3.63	0.27
gpt-4o	2.85	2.76	2.93	0.09
gpt-4o-mini	2.10	2.25	1.60	0.34

The factual quarter is where the runs differ. On the twelve-evaluator basis, with the consistency ratings from run 1’s sweep throughout:

Table 15: Factual competence E^F per run and pooled (twelve-evaluator basis). Spectral gap σ_1/σ_2 : 2.57, 1.67, 1.29; pooled 2.24 (runs 1 and 2 alone, 2.45). Run-to-run agreement (Pearson) of E^F alone: 0.90, 0.79, 0.80 for runs 1–2, 1–3, 2–3; of the three regenerated quarters $\frac{1}{3}(G^F + G^C + E^F)$: 0.95, 0.91, 0.93; of T : 0.97, 0.94, 0.95.

Model	E^F run 1	run 2	run 3	pooled
gemini-3.1-pro	7.36	8.29	10.24	8.24
$\star$ claude-opus-4.5	7.00	7.00	7.00	7.00
claude-opus-4.0	4.47	3.48	5.41	4.69
claude-opus-4.1	3.55	6.78	5.17	4.65
gemini-2.5-flash	4.68	3.81	0.01	3.47
claude-sonnet-4	1.62	2.44	4.00	2.37
gpt-4.1-mini	1.21	3.88	0.06	1.34
gpt-4.1-2025-04-14	0.20	0.53	2.92	0.79
gpt-4o	0.34	0.41	1.02	0.47
gpt-4o-2024-08-06	0.62	0.00	0.00	0.35
gpt-4.1-nano	0.49	0.39	0.00	0.28
gpt-4o-mini	0.00	0.00	0.00	0.00

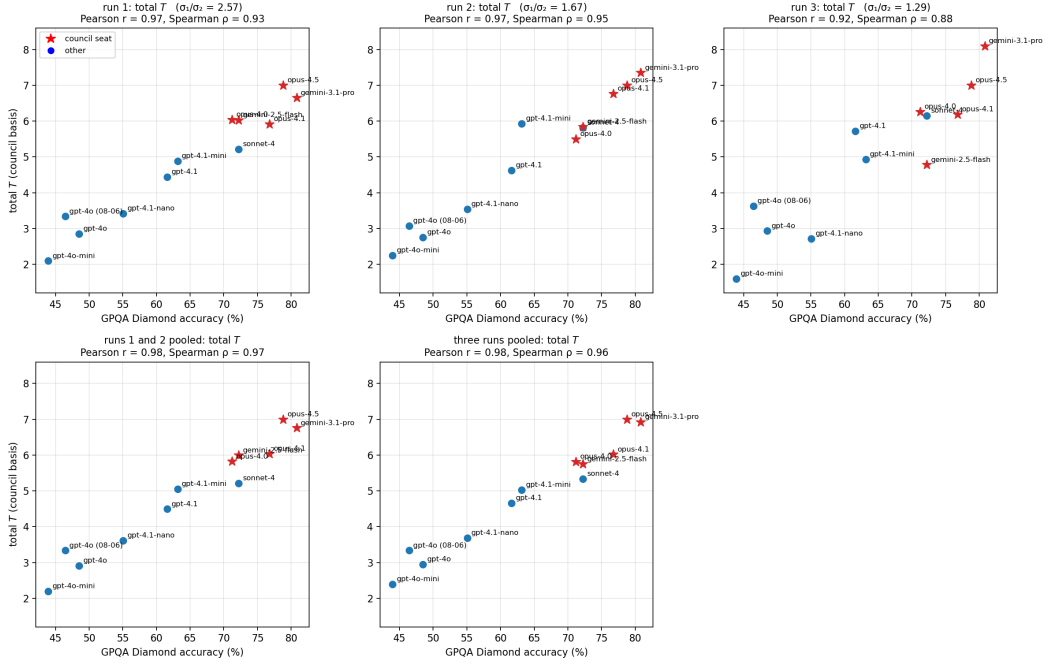


Figure 4: The three runs apart and pooled, against GPQA Diamond. Top row: council-basis T per run, with each run’s twelve-evaluator spectral gap. Bottom row: runs 1 and 2 pooled (left), all three pooled (right). Run 3, with the smallest gap, is the odd one out; pooling restores an identified factual axis and the agreement rises.

σ_1/σ_2 is the standard spectral-gap diagnostic for whether a leading singular direction is identified (Davis–Kahan: the rotation of u_1 under perturbation is bounded by the gap to σ_2 and by nothing below it); the threshold of 2 is a rule of thumb, not a derived value. Run 2 sits below it and reads well; run 3 sits further below and does not. On this evidence a threshold near 1.5 may separate the two, but two runs cannot fix it, and GPQA, the external reference that names run 3 as the odd one out, is not ground truth for intelligence and sets no protocol parameter here. The benchmark’s policy is therefore to pool: three portfolios per player, the seats’ and the ballast’s three stored, one factorisation over all of them, and the guard reported on the pooled matrix.

G What anchoring does to resolution

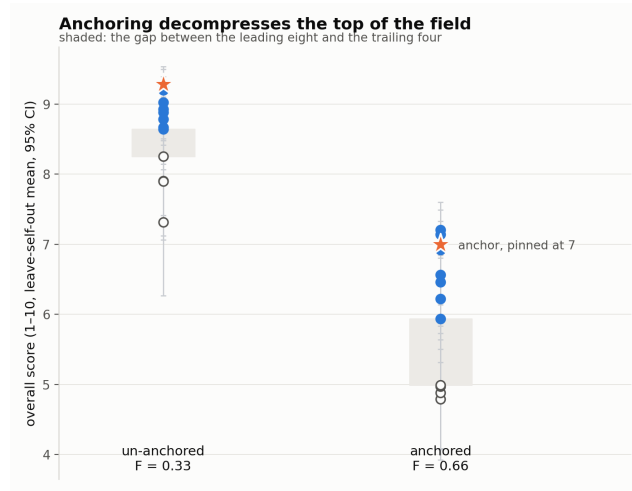


Figure 5: Each model’s leave-self-out overall mean with its 95% bootstrap CI, un-anchored (left) and anchored (right); models are unlabelled deliberately — the message is structural, not a ranking. Filled markers are the leading eight, open the trailing four, the shaded band the gap between them; the star is the anchor — a scored target on the left, the pinned reference (7) on the right. Anchored scores spread around the yardstick instead of piling against the ceiling: the gap more than doubles relative to the spread of the means (every scored cross-band pair holds at bootstrap probability 1.00), the variance ratio — between-target over within-target variance, Fisher’s F — doubles from 0.33 to 0.66, and the intervals still overlap within each band.

H The argument behind the hypothesis

Nothing in this appendix tests the hypothesis of §6. It sets out the argument step by step, marks each step as a cited result, a measurement on the released data, or the hypothesis itself, and ends with what would refute it. Every number comes from a script in the released package.

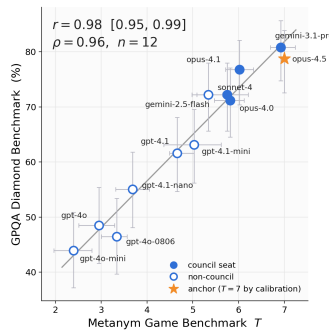


Figure 6: The agreement the hypothesis is offered for, reproduced from Figure 2: the official total T (council basis, three runs pooled) against self-administered GPQA Diamond accuracy, twelve models; Pearson $r = 0.98$ (95% BCa CI [0.95, 0.99]), Spearman $\rho = 0.96$. Filled markers are council seats, open markers non-council, the star the anchor at $T = 7$ by calibration.

H.1 The chain

1. Training a language model is compression (cited). Predicting text is compressing it (Shannon, 1951); a trained model is a general-purpose compressor (Delétang et al., 2024).
2. Training selects what is compressed (cited, by construction of the objective). The loss is taken on held-out text, so an abstraction earns its place only by predicting instantiations the model has not seen, and the abstractions that recur most across the corpus earn the most. A situation structure that appears in biology, economics and stories alike is one.
3. Compression generalises (cited). The shortest description keeps a recurring pattern once, as an abstraction, and reuses it (Rissanen, 1978); that is a concept.
4. A concept is a frame with open slots (cited). Minsky’s (1974) frames are stereotyped situations with terminals that instances fill; the context template of §2 is such a frame, written in words.
5. An analogy is a shared relational structure with different particulars (cited; Gentner, 1983). The archetypal context is that shared structure.
6. Hypothesis. A language model holds its knowledge as archetypal contexts, each instantiated in the topic domains where it applies. The template is the abstraction; the metonym set is what the abstraction leaves out, the topic domain; the two reconstruct the instantiation exactly.
7. The metonym form is strong compression (measured, H.2). About 3 for a template with five parallel contexts and about 8 for each further one, growing with every domain a template serves — the order of the compression a trained model achieves on text, in words rather than bits. The factor grows because the template has open slots: a memorised string compresses only its exact repeats, a frame compresses every new instance that fits it, and no instantiation here repeats another. This is the abstracting kind of compression, the kind held-out loss rewards; a representation this compressive is what training would find.
8. The game measures the mechanism directly (hypothesis). Nothing is given; the model selects the archetype and the domains that best satisfy the prompt’s criteria, factual truth among them, and writes them out.
9. GPQA measures the mechanism’s use (hypothesis, with H.4 as the observation). The archetype and the domain are given by the question; the model derives the instantiation and selects the candidate that matches it. The two tests agree because they exercise one operation from opposite ends: GPQA fixes the store position and selects among four strings, the game selects the store position and fixes the output form.
10. Corollary: retrieval, not construction (hypothesis, with H.3 as the signature). In the game the models retrieve templates and metonym sets that training already compressed, which is why they play well with no reasoning channel and why a model rewrites its portfolios between runs but keeps its archetypes.
11. What the data establish is narrower. The factual pair reaches 0.94 with GPQA and the two subjective quarters lift it to 0.98 (§4.5, Appendix D.1). The hypothesis is consistent with the general factor of Ilić and Gignac (2024), and more specific.

H.2 The compression of the metonym form

Measured on the anchor submission’s first archetype, the one archetype shipped in full (scripts/compression_ratio.py). The template is 120 words; the metonym table 84 words for five parallel contexts, 17 per context; the five instantiations (Form a, the form the template and metonym set reconstruct exactly) average 130 words. With n contexts the factor is $130n/(120 + 17n)$:

Table 16: Literal compression of the metonym form against the number of parallel contexts per template, from the anchor’s first archetype (120-word template, 17 words of metonyms per context, 130-word instantiations).

contexts	5	10	20	50	limit
factor	3.2	4.5	5.7	6.9	7.8

The asymptote is the ratio of an instantiation to its metonym set, and it is reached only when a template is amortised over dozens of domains, which is the situation inside a model rather than inside a five-context game. The count is in words, and metonyms are the high-entropy tokens, so the factor in bits is lower; the bit-level factor is measurable with an open-weights model and is not measured here. Templates in the released portfolios run 80 to 120 words with 10 to 15 slots, so the asymptote varies by archetype within the same range. It is a lower limit in one further sense: a metonym set is itself a frame with slots, so the topic domain compresses by the same rule, and the factors multiply down the hierarchy.

H.3 The retrieval signature in the released transcripts

The archetype titles and parallel-context domains of every graded portfolio in the three runs, as the evaluation transcripts record them (scripts/archetype_recurrence.py; the raw portfolios are not shipped, the transcripts copy their headings).

Within a model, the archetypes persist while the texts change. Eleven of twelve portfolios differ between runs (§4.6). Between the two runs made two hours apart, 34 of 48 archetype titles recur verbatim; between run 1 and run 2, three weeks apart, 20 of 50. Gemini 2.5 Flash produced the same five titles in all three runs; Claude Opus 4.0 kept four or five of five. The models rewrite; they do not reinvent. This is what one archetype in latent space with many projections in literal space predicts, and what a stored text would not.

Across models, the same concepts recur, across vendors. Seven of the eleven graded models, from all three vendors, offer a resource-allocation archetype (title containing “allocation”). Six domains are used by five or more models: urban planning, social media, corporate structure, the climate system, software development, biological evolution. Of 81 distinct titles, 14 are used by more than one model.

The caveat. Recurrence is strongest inside a vendor: 11 of the 14 shared titles are shared within one vendor only, the two Opus models sharing cascade amplification, phase transition, competitive exclusion and boundary maintenance word for word, and the GPT-4o family sharing resource allocation, conflict resolution and growth and development. Shared training data explains within-vendor recurrence as well as shared representation does. The cross-vendor recurrence is the evidence.

H.4 How GPQA was answered

With the reasoning channel off and temperature zero, no model answered with a bare letter (scripts/gpqa_reply_lengths.py):

Table 17: Words per GPQA reply with the reasoning channel off, and replies under fifteen words, per model (198 items each).

model	median words	replies under 15 words
claude-opus-4.0 / 4.1 / 4.5	221 / 242 / 288	2 / 2 / 0
claude-sonnet-4	260	0
gemini-2.5-flash / 3.1-pro	546 / 374	0 / 2
gpt-4.1 / mini / nano	180 / 328 / 287	5 / 0 / 0
gpt-4o / 2024-08-06 / mini	196 / 191 / 230	1 / 1 / 0

Thirteen replies of 2,376 are under fifteen words, most of them truncations. Every model wrote a derivation and ended it with the answer line, as the prompt allowed (Appendix D.2). In the derivations the model states the relation the question turns on in the question’s domain and then tests the candidates against it: an instantiation written out, then a match — the sequence step 9 describes. “Reasoning off” removes the hidden channel; the visible derivation remains, and the question of whether a hidden channel would change play is open (§6, Scope).

H.5 Predictions, and what would refute the hypothesis

1. Domain-matched agreement. A model’s factual score on parallel contexts in a field tracks its GPQA accuracy in that field. GPQA Diamond carries subject labels, and the released evaluations carry each parallel context’s domain, so the test needs no new run. A refutation: the field-by-field agreement is no higher than the agreement across fields.
2. Separability in latent space. Archetype and topic domain are separable in a model’s latent space: the same archetype recoverable from parallel contexts in unrelated domains, and the domain from unrelated archetypes. A refutation: representations that separate by domain only.
3. Retrieval over construction. Reasoning should add little to generation in the game, where templates and metonym sets are retrieved, and may still help judging, where the task is to catch a broken substitution. A refutation: a reasoning budget that raises generation ratings as much as it raises judging.
4. The bit-level factor. Given its template and metonym set, an instantiation’s per-token loss under an open-weights model should fall far below its loss given nothing, and by more for archetypes that recur across many domains. A refutation: no such gap, or a gap that does not grow with the number of domains.